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Inventor(s)	Sanjeevappa; Manjunatha Kaankanhalli et al.

Suction manifold assembly for an electrocardiogram electrode application system

Abstract

An ECG electrode application system includes an electrode distributor including a housing and a suction manifold assembly disposed within the housing. The suction manifold assembly includes a manifold, a plurality of pneumatic valves integrated within the manifold, and the manifold includes a common manifold chamber fluidly coupled to each pneumatic valve, and each pneumatic valve includes a venturi passage. The ECG electrode application system includes a plurality of suction electrodes, each suction electrode having an electrode suction dome and coupled to the electrode distributor via a lead wire, and each lead wire is coupled to a respective pneumatic valve. Each electrode suction dome is configured when pressed to cause a change in air pressure that results in a suction pressure being applied within a respective electrode suction dome to adhere a corresponding suction electrode to a subject at a pressure determined by the venturi passage of a corresponding pneumatic valve.

Inventors: **Sanjeevappa; Manjunatha Kaankanhalli (Bangalore, IN), Ippili; Gangadhar (Bangalore, IN), Issac; Robin Mathew (Bangalore, IN), Sivasankaran; Sirosh (Bangalore, IN), Varghese; Oswin (Bangalore, IN), Filtz; Sharla Rae (Cedarburg, WI), Strobel; Caitlin Theresa (New Castle, PA)**

Applicant: **GE Precision Healthcare LLC** (Wauwatosa, WI)

Family ID: **88513632**

Assignee: **GE Precision Healthcare LLC (Wauwatosa, WI)**

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Primary Examiner: Dvorak; Linda C

Assistant Examiner: Mossbrook; William

Attorney, Agent or Firm: Fletcher Yoder, P.C.

Background/Summary

BACKGROUND

(1) The subject matter disclosed herein relates to an electrocardiogram (ECG) electrode application system.

(2) A variety of physiological parameters may be measured by monitoring electrical signals within the body. Monitoring systems utilizing electrodes may be utilized to monitor the transfer of electrical energy within the body of a patient. For example, a monitoring system may include an ECG monitoring system that utilizes electrodes (e.g., ECG electrodes) to monitor heart activity during various states (e.g., resting and/or exercise). Some ECG monitoring systems may utilize electrodes (e.g., suction electrodes) that are adhered to the patient's skin via negative pressure (e.g., due to a vacuum). In certain aspects, these suction electrodes may be adhered via negative pressure to the patient in response to a trigger (e.g., pressing a dome associated with the suction electrode). However, the sensitivity of the trigger (e.g., electrode dome) may require a large amount of force to trigger the negative pressure. Sometimes an electrode dome may need to be pressed multiple times to trigger the negative pressure. In addition, the negative pressure (e.g., suction pressure) may be too little resulting in failure to acquire an ECG signal via the electrode due to insufficient contact of the electrode with the patient. Also, the negative pressure may be too much resulting in a wound or cut on the skin of the patient.

BRIEF DESCRIPTION

(3) Certain aspects commensurate in scope with the originally claimed subject matter are summarized below. These aspects are not intended to limit the scope of the claimed subject matter, but rather these aspects are intended only to provide a brief summary of possible aspects. Indeed, the subject matter may encompass a variety of forms that may be similar to or different from the aspects set forth below.

(4) In one aspect, an electrocardiogram (ECG) electrode application system is provided. The ECG electrode application system includes an electrode distributor. The electrode distributor includes a housing and a suction manifold assembly disposed within the housing. The suction manifold assembly includes a manifold, a plurality of pneumatic valves integrated within the manifold, and the manifold includes a common manifold chamber fluidly coupled to each pneumatic valve of the plurality of pneumatic valves, and each pneumatic valve includes a venturi passage. The ECG electrode application system also includes a plurality of suction electrodes, each suction electrode of the plurality of suction electrodes having an electrode suction dome and coupled to the electrode distributor via a lead wire, and each lead wire is coupled to a respective pneumatic valve. Each electrode suction dome is configured when pressed to cause a change in air pressure that results in a suction pressure being applied within a respective electrode suction dome to adhere a corresponding suction electrode to a subject at a pressure determined by the venturi passage of a corresponding pneumatic valve.

(5) In another aspect, a suction manifold assembly for an ECG electrode application system is provided. The suction manifold assembly includes a manifold. The suction manifold assembly also includes a plurality of pneumatic valves integrated within the manifold. The suction manifold assembly also includes a common manifold chamber fluidly coupled to each pneumatic valve of the plurality of pneumatic valves, and wherein each pneumatic valve is configured to be coupled via a lead wire to a respective suction electrode. The suction manifold assembly is configured to be disposed within a housing of an electrode distributor of the ECG application system. The pneumatic valves of the plurality of pneumatic valves are disposed adjacent to each other along a horizontal plane in both a first direction and a second direction, the first direction being orthogonal to the second direction.

(6) In a further aspect, an ECG electrode application system is provided. The ECG electrode application system includes an electrode distributor. The electrode distributor includes a housing and a suction manifold assembly disposed within the housing. The suction manifold assembly includes a manifold, a plurality of pneumatic valves integrated within the manifold, and the manifold includes a common manifold chamber fluidly coupled to each pneumatic valve of the plurality of pneumatic valves. The ECG electrode system also includes an orifice disposed within the common manifold chamber. The ECG electrode application system further includes a plurality

of suction electrodes, each suction electrode of the plurality of suction electrodes having an electrode suction dome and coupled to the electrode distributor via a lead wire, and each lead wire is coupled to a respective pneumatic valve. Each electrode suction dome is configured when pressed to cause a change in air pressure that results in a suction pressure being applied within a respective electrode suction dome to adhere a corresponding suction electrode to a subject at a pressure determined by the venturi passage of a corresponding pneumatic valve, and the orifice is configured to set a level of change in the air pressure to trigger the application of the suction pressure applied within a respective electrode suction dome.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and other features, aspects, and advantages of the present subject matter will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:

(2) FIG. **1** is a schematic diagram of an ECG electrode application system, in accordance with aspects of the present disclosure;

(3) FIG. **2** is a top view of an electrode distributor coupled to a regulator of the ECG electrode application system in FIG. **1**, in accordance with aspects of the present disclosure;

(4) FIG. **3** is a side schematic view of pneumatic valves of a manifold of the electrode application system in FIG. **1** relative to horizontal planes, in accordance with aspects of the present disclosure;

(5) FIG. **4** is a perspective view of a plurality of suction electrodes of the electrode application system in FIG. **1** coupled to respective leads, in accordance with aspects of the present disclosure;

(6) FIG. **5** is a perspective view of a bottom surface (e.g., patient interfacing surface) of a suction electrode of FIG. **4**, in accordance with aspects of the present disclosure;

(7) FIG. **6** is a perspective view of a portion of a pneumatic valve of the electrode system in FIG. **1** defined by a manifold, in accordance with aspects of the present disclosure;

(8) FIG. **7** is a cross-sectional view of the portion of the pneumatic valve in FIG. **6**, in accordance with aspects of the present disclosure;

(9) FIG. **8** is a perspective of a pneumatic valve of the electrode system in FIG. **1** coupled to a common manifold chamber, in accordance with aspects of the present disclosure;

(10) FIG. **9** illustrates the velocity of the air flow through a venturi path of the pneumatic valve in FIGS. **6-8** (e.g., via CFD modeling), in accordance with aspects of the present disclosure;

(11) FIG. **10** is a perspective view of a regulator having an orifice of the ECG electrode application system in FIG. **1**, in accordance with aspects of the present disclosure;

(12) FIG. **11** is a cross-sectional view of the regulator having the orifice in FIG. **10** and a portion of an electrode distributor coupled to an electrode distributor, in accordance with aspects of the present disclosure;

(13) FIG. **12** is a partial cut-away view of an electrode distributor coupled to a regulator, in accordance with aspects of the present disclosure;

(14) FIG. **13** is a cross-sectional view of a pneumatic valve of the ECG electrode application system in FIG. **1** (e.g., in a presence of a change in pressure from pressing a suction electrode), taken along line **13-13** of FIG. **8**, in accordance with aspects of the present disclosure;

(15) FIG. **14** is a cross-sectional view of the pneumatic valve in FIG. **14** (e.g., in the presence of a vacuum), taken along line **13-13** of FIG. **8**, in accordance with aspects of the present disclosure;

(16) FIG. **15** is a side view of an electrode distributor of the ECG electrode application system in FIG. **1** coupled to lead wires, in accordance with aspects of the present disclosure; and

(17) FIG. **16** is a perspective view of the ECG electrode application system in FIG. **1**, in

accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

(18) One or more specific aspects will be described below. In an effort to provide a concise description of these aspects, not all features of an actual implementation are described in the specification. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be appreciated that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

(19) When introducing elements of various aspects of the present subject matter, the articles “a,” “an,” “the,” and “said” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Furthermore, any numerical examples in the following discussion are intended to be non-limiting, and thus additional numerical values, ranges, and percentages are within the scope of the disclosed aspects.

(20) The present disclosure provides for an ECG electrode application system to monitor ECG activity of a patient in various states (e.g., during and exercise). In particular, the ECG electrode application system may include an electrode distributor that includes a housing that includes a suction manifold assembly disposed within. Leads of suction electrodes (e.g., suction ECG electrodes) may be coupled to a bottom surface of the electrode distributor. In certain aspects, these leads may extend in a vertical direction from the bottom surface of the electrode distribution system to eliminate or minimize bending stresses in the leads, thus, improving the life spans of the leads. The suction manifold assembly includes a manifold. The manifold includes a plurality of pneumatic valves integrated within the manifold. Adjacent pneumatic valves may be arranged in both a first direction and a second direction (which are orthogonal with respect to each other) along a horizontal plane. Each pneumatic valve is fluidly coupled to a common manifold chamber of the manifold. Each lead wire is fluidly coupled to a corresponding pneumatic valve. Each pneumatic valve includes a venturi path integrated within the manifold. The venturi path determines the negative pressure (e.g., suction dome pressure) with which the suction electrode is adhered to the skin of a patient. An orifice is disposed with the common manifold chamber. The orifice determines the sensitivity (e.g., by controlling mass flow rate within the system) for pressing the suction dome to trigger a pump (e.g., disposed within the electrode distributor) to cause a vacuum to create the negative pressure for adhering the suction electrodes to the skin of the patient. The utilization of the venturi paths enables a compact manifold. The combination of the venturi paths and the orifice improves triggering the use of a pump and provides sufficient vacuum transfer pressure to the suction electrodes.

(21) FIG. **1** is a schematic diagram of an ECG electrode application system **10** according to an example. The ECG electrode application system **10** enables ECG activity to be acquired and monitored for a patient in various states (e.g., during and exercise). The ECG electrode application system **10** includes an electrode distributor **12** having a housing **14**. The electrode distributor **12** includes a suction manifold assembly **16** disposed within the housing **14**. The suction manifold assembly **16** includes a manifold **18** coupled to a lead wire printed circuit board (PCB) **20**. The electrode distributor **12** also includes a pump **22** (e.g., suction or vacuum pump) and a control board **24** (e.g., pressure sensor board) for the pump **22** disposed within the housing **14**. The pump **22** is coupled to the manifold **18** via tubing. A regulator **26** (e.g., vacuum regulator) is coupled to an end **28** of the manifold **18** opposite an end **30** where the pump **22** is located.

(22) The suction manifold assembly **16** includes a plurality of pneumatic valves **32** (e.g., suction lead valves) integrated within the manifold **18**. The number of pneumatic valves **32** integrated

within the manifold **18** may vary. As depicted, the number of pneumatic valves **32** is **14**. In certain aspects, the number of pneumatic valves **32** may be **10**, **12**, or another number. As described in greater detail below, adjacent pneumatic valves **32** may be arranged in a first direction and a second direction (that are orthogonal with respect to each other) along a horizontal plane (e.g., extending form end **28** to end **30** and from side **29** to side **31**). Each pneumatic valve **32** includes a venturi path or passage **34**. A portion of each pneumatic valve **32** (including the venturi path **34**) may be additively manufactured as part of the manifold **18**. The manifold **18** includes a common manifold chamber which is fluidly coupled to each pneumatic valve **32**. The venturi paths **34** enable the manifold **18** to be more compact compared to other manifolds associated with other ECG electrode application systems.

(23) The ECG electrode application system **10** includes suction electrodes **36** (e.g., ECG electrodes) that are coupled to the lead wires **38**. Each lead wire **38** of the suction electrodes **38** is configured to couple (e.g., via a port connector) to a surface (e.g., bottom surface) of the electrode distributor **12** via ports. Each lead wire **38** includes pneumatic tubing and may be made of a conductive plastic. When coupled to the ports, the lead wires are coupled to and provide to the lead wire PCB **20** electrical signals (ECG signals) from the suction electrodes **36**. When coupled to the ports, the pneumatic tubing of each lead wire **38** is fluidly coupled to a respective manifold port associated with a respective pneumatic valve **32**.

(24) The ECG electrode application system **10** includes an area or provision **40** for holding and coupling an acquisition module **42** to the electrode distributor **12** (e.g., electrode block distributor). The electrode distributor **12** includes a PCB board that couples to the acquisition module **42** when the acquisition module is coupled to the electrode distributor **12**. Electrical signals from the electrode sensors are passed from the lead wire PCB **20** to the PCB board coupled to the acquisition module **42** and then to the acquisition module **42**. The acquisition module **42** acquires the analog ECG signals from the suction electrodes **36**, digitizes them, and sends them to a host system. The area or provision **40** also enables a trunk cable (e.g., having a pneumatic tubing connector) to be coupled to the lead wires **38** via the suction manifold assembly **16**. The ECG electrode application system **10** receives power from a power supply (not shown).

(25) The lead wires **38**, the suction manifold assembly **16**, the regulator **26**, and the control board **24**, and the pump **22** form a system that enables air to flow at a certain mass flow rate (e.g., 3.04×10.sup.5 kg/s) and air velocity. The vacuum regulator **26** is coupled to an orifice **44** that extends into manifold **18**. An inner diameter of the orifice **44** determines a level of change in air pressure (i.e., the sensitivity of the electrode dome of each suction electrode **36**) that will trigger the pump **22** to create a vacuum or suction pressure in the system. The orifice **44** minimizes the number of presses of the suction dome of a respective suction electrode **36**. The regulator **26** is configured to alter the amount of vacuum or suction pressured applied (e.g., between approximately 150 to 195 mmHg).

(26) Each suction electrode **36** includes an electrode (e.g., silver/silver chloride electrode) and a suction dome (e.g., made of a conductive plastic) disposed about the electrode. Each suction electrode **36** may also include a filter disk. To apply the suction electrode **36** to the skin of a patient, the suction dome is pressed and released (e.g., via a finger) which generates a small air pressure or change in air pressure (e.g., approximately 5 to 10 mmHg) in the system. In particular, pressing the suction dome compresses and pressurizes the air in the system. The air pressure actuates the pneumatic valve **32** associated with the suction electrode **36** and opens the venturi passage **34** to the air path. The small, pressurized air passes through the lead wires **38**, the manifold **18** (including the pneumatic valve **32**), and the regulator **26** to the pressure sensor on the control board **24**. The control board **24** senses the change in air pressure and triggers the pump **22** for actuation. Upon the vacuum being applied in the system, the suction electrode **36** adheres and holds its place on the skin of the patient. The suction dome pressure of the suction electrode **36** adhering to the patient is determined by the venturi path **34**. The venturi path **34** provides the proper required pressure to

adhere the suction electrode **36** to acquire an ECG signal without wounding or cutting the skin of the patient. The venturi path **34** also increases the mass flow of air in the system, which is controlled by the orifice **44**.

(27) FIG. **2** is a top view of an electrode distributor **12** coupled to the regulator **26** of the ECG electrode application system **10** in FIG. **1**, according to an example. The electrode distributor **12** includes a wall **46** defining the housing **14**. The wall **46** defines a bottom surface (not shown, see FIG. **13**), the end **30**, and the sides **29** and **31**. The regulator **26** is coupled to end **28** of the electrode distributor **12** to close it off. A lid (not shown, see FIG. **12**) couples to the electrode distributor **12** to enclose the internal components (e.g., the pump **22**, the suction manifold assembly **16**, etc.) within the electrode distributor **12**. As depicted in FIG. **2**, a PCB board **48** includes connections **50** for coupling to an acquisition module (e.g., acquisition module **42** in FIG. **1**).

(28) As depicted in FIG. **2**, the manifold **18** is disposed on and coupled to the lead wire printed circuit board (PCB) **20**. The lead wire PCB **20** is electrically coupled to the lead wires of the suction electrodes. The plurality of pneumatic valves **32** are integrated within the manifold **18**. As depicted, the pneumatic valves **32** are disposed adjacent to each other along a horizontal plane in a first direction **52** (e.g., extending from end **28** to end **30**) and second direction **54** extending from side **29** to side **31**. The first direction **52** is orthogonal to the second direction **54**. In particular, a first set **56** of the pneumatic valves **32** extend in both the first direction **52** and the second direction **54** along a horizontal plane **58** (see FIG. **3**). A second set **60** of the pneumatic valves **32** extend in both the first direction **52** and the second direction **54** along another horizontal plane **62** (see FIG. **3**) that is different from and parallel with the horizontal plane **58**. The first set **56** of the pneumatic valves **32** is located above the second set **60** of pneumatic valves **32** in direction **64** (which is orthogonal with respect both the first direction **52** and the second direction **54**). In particular, first set **56** of the pneumatic valves **32** is disposed more adjacent to the top surface (i.e., the lid) of the housing **14** of the electrode distributor **12**, while the second set **60** of pneumatic valves **32** is disposed more adjacent to the bottom surface of the housing **14**. Some of the pneumatic valves **32** in the first set **56** are vertically aligned with (along the direction **64**) with some of the pneumatic valves **32** in the second set **60**. The first set **56** of the pneumatic valves **32** and the second set of pneumatic valves **32** are oriented in opposite directions as depicted in FIG. **2**. In particular, a retainer ring (see FIGS. **13** and **14**) of each pneumatic valve **32** of the first set **56** faces the top surface (i.e., the lid) of the housing **14**, while the retainer ring of each pneumatic valve **32** of the second set **60** faces the bottom surface of the housing **14**. As depicted, the first set **56** of pneumatic valves **32** includes 6 pneumatic valves **32** and the second set **60** of pneumatic valves **32** includes 8 pneumatic valves **32**.

(29) As depicted in FIG. **2**, tubing **66** connects the pump **22** to the manifold **18**. In addition, the tubing **66** connects manifold **18** to the control board **24** having the pressure sensor. The control board **24** is also connected to the pump **22**. The electrode distributor **12** also includes a connection **68** for coupling to a trunk cable **70** (e.g., having a pneumatic tubing connector).

(30) FIGS. **4** and **5** illustrate different views of the suction electrodes **36** (e.g., ECG electrodes), according to an example. Each suction electrode **36** includes an electrode **72** (e.g., silver/silver chloride electrode) and a suction dome **74** (e.g., made of a conductive plastic) disposed about the electrode **72**. Each suction electrode **36** may also include a filter disk. To apply the suction electrode **36** to the skin of a patient, a top portion **76** the suction dome **74** is pressed and released (e.g., via a finger) which generates a small air pressure or change in air pressure (e.g., negative air pressure) (e.g., approximately 5 to 10 mmHg) in the system. In particular, pressing the suction dome **74** compresses and pressurizes the air in the system. As depicted in FIG. **4**, each suction electrode **36** is coupled to the lead wire **38** (e.g., having pneumatic tubing and made of conductive plastic). Each suction electrode **36** is coupled to the lead wire **38** via a connector **78**.

(31) FIGS. **6** and **7** are different views of a portion of the pneumatic valve **32** defined by the walls of the manifold (e.g., manifold **18** in FIG. **1**), according to an example. The pneumatic valve **32**

includes an inlet portion **80** (e.g., manifold port portion) and an outlet portion **82** (e.g., forming the venturi path **34**) coupled to a valve body portion **84** (e.g., central portion). The inlet portion **80**, the outlet portion **82**, and the valve body portion **84** are integral to the manifold. In particular, inlet portion **80**, the outlet portion **82**, and the valve body portion **84** may be additively manufactured (e.g., via stereolithography) as part of the manifold. After additive manufacturing, the additively manufactured portion of the pneumatic valve has pressurized air mixed with iso-propyl alcohol passed through (e.g., sometimes repeatedly) the manifold path to evacuate resins stuck in the air path to achieve the desired air pressure difference between the start and end of the air path.

(32) The inlet portion **80** is fluidly coupled to the lead wire of a suction electrode. The outlet portion **82** is fluidly coupled to a common chamber (e.g., vacuum chamber) of the manifold. As described in greater detail below, the valve body portion **84** is configured to receive a plunger within, where the plunger changes positions relative to the valve body portion in response to pressure changes. A diameter **86** (see FIG. **7**) of the inlet portion **80** narrows or tapers as it approaches the valve body portion **84**. The valve body portion **80** includes a narrow portion **86** coupled to the outlet portion **82**. The valve body portion **80** includes a diameter **90** adjacent the narrow portion **88**. The narrow portion **86** includes a diameter **92**. The outlet portion **82** includes a diameter **94** along its length. The diameter **92** is less than the diameter **90**. The diameter **94** is less than the diameter **90** and greater than the diameter **92**. The narrower diameter **92** in the narrow portion **88** creates a venturi effect in the flow of air from the valve body portion **80** to the outlet portion **82** (e.g., the venturi path **34**). In certain aspects, the diameter **94** is approximately 2 millimeters (mm). In certain aspects, the diameter **92** is approximately 0.75 mm. In certain aspects, the diameter **92** may another value (e.g., approximately 0.64 mm or 1 mm). In certain aspects, the venturi path **34** may have a length of approximately 1.6 mm.

(33) FIG. **8** illustrates the inlet portion **80**, the outlet portion **82**, and the valve body portion **84** integrated within a portion of the manifold **18**, according to an example. As depicted in FIG. **8**, the outlet portion **82** or venturi path **34** extends into a common manifold chamber **96**. As depicted in FIG. **8**, the orifice **44** is extending into the common manifold chamber **96**.

(34) FIG. **9** illustrates the velocity of the air flow through the venturi path **34** of the pneumatic valve **32** in FIGS. **6-8** (e.g., where the narrow portion has a diameter of approximately 0.75 mm). The velocity of air flow is fairly low through the inlet portion (e.g. inlet portion **80** of FIG. **8**) and the valve body portion **84** of the pneumatic valve **32**. As depicted in FIG. **9**, the velocity of the air flow significantly increases as it enters (e.g., via the narrow portion **88**) the venturi path **34** or the outlet portion **82** from which it eventually exits. The velocity of the air flow is at its highest in narrow portion **88** as it enters the outlet portion **84** (e.g., due to the narrower diameter of the narrow portion **88**). The velocity of the air flow is significantly higher in the outlet portion **84** than the inlet portion (e.g., inlet portion **80**). The average inlet velocity of the air flow (e.g., into the inlet portion) is approximately 4.62 m/s. The average outlet velocity of the air flow (e.g., from the outlet portion **84**) is approximately 30.313 m/s. The mass flow rate at outlet from the outlet portion **82** is approximately 8.09×10.sup.5 kg/s.

(35) FIG. **10** is a perspective view of the regulator **26** (e.g., vacuum regulator) having the orifice **44**. The orifice **44** is coupled to an inner surface **98** (e.g., surface that faces an interior of an electrode distributor). The orifice **44** extends away from the surface **98** toward an interior of an electrode distributor. As depicted in FIG. **11**, the regulator **26** is coupled the end **28** of the housing **14** of the electrode distributor **12**. A lid **100** (e.g., forming a top surface) is disposed on the housing **14** and the regulator **26**. The manifold **18** includes a lower chamber **102** and the common manifold chamber **96**. The common manifold chamber **96** is fluidly coupled to the outlet portion **82** or venturi path **34** associated with each pneumatic valve. The lower chamber **102** and the common manifold chamber **96** are fluidly coupled to each other via the regulator **26**. In particular, air flows between chamber **102** and the common manifold chamber **96** via the regulator **26**. As depicted, the orifice **44** extends from the regulator **26** into the common manifold chamber **96**. The orifice **44**

includes an inner diameter **104**. In certain aspects, the inner diameter **104** may be approximately 1.198 mm. The venturi air paths **34** increase the mass flow rate of air in the system and determine the suction pressure applied by each suction electrode. The inner diameter **104** of the orifice **44** controls the mass flow rate of the air in the system as well as the sensitivity of the pressing of the electrode dome of each suction electrode (i.e., the change in pressure needed to trigger the vacuum or suction pressure created by the pump).

(36) FIG. **12** is a partial cut-away view of the electrode distributor **12** coupled to the regulator **26**. As depicted, the pump **22**, the control board **24**, and the suction manifold assembly **16** are disposed within the housing **14** of the electrode distributor **12**. The lead wire PCB is not shown. The regulator **26** is coupled to the end **28**. Tubing **66** connects the pump **22** to the manifold **18**. In addition, the tubing **66** connects manifold **18** to the control board **24** having the pressure sensor. The control board **24** is also connected to the pump **22**. The electrode distributor **12** also includes a connection **68** for coupling to a trunk cable **70** (e.g., having a pneumatic tubing connector).

(37) As depicted in FIG. **12**, the lead wires **38** of the suction electrodes are coupled to a bottom surface **106** of the housing **14**. In particular, p-connectors **108** are disposed in ports in the bottom surface **106**. The p-connectors **108** extend vertically in direction **64** from the bottom surface **106**. This minimizes or eliminates bending stresses in the lead wires **38**, thus, improving the life span of the lead wires **38**.

(38) In the absence of triggering a vacuum via pressing a suction electrode, air flows (as indicated by arrows **110**) through the tubing **66** and into the lower chamber **102** of the manifold **18**. From the lower chamber **102**, the air flows through the regulator **26** into the common manifold chamber **96**. Air then flows from the common manifold chamber **96** into the pneumatic valves **32** via the venturi paths **34** as indicated by arrows **112**. Then, the air flows through the pneumatic valves and exits the portions **80** (as indicated by arrows **114**) to the lead wires **38** of the suction electrodes. Overall due to the orifice and the venturi paths **34**, in certain aspects, the mass flow rate in the system is approximately 3.036×10.sup.5 kg/s.

(39) FIGS. **13** and **14** are cross-sectional views of the pneumatic valve **32** taken along line **13-13** of FIG. **8**. The pneumatic valve **32** includes the inlet portion **80**, the outlet portion **82**, and the valve body portion **84** integrated within a portion of the manifold **18** as described above. The pneumatic valve **32** also includes a plunger **116** disposed within the valve body portion. The plunger **116** includes a first end portion **118** (e.g., flat portion) and a second end portion **120**. An O-ring **122** is disposed about the second end portion **120**. A spring **124** is disposed about the second end portion **120** on a side of the O-ring **122** opposite the first end portion **118**. A retainer ring **126** is secured to the manifold **18** to keep the plunger **116** within the pneumatic valve **32**. A paper barrier **128** is disposed within the pneumatic valve **32** between the first end portion **118** and the retainer ring **126**.

(40) FIG. **13** depicts the pneumatic valve **32** in a presence of a change in pressure from pressing the suction electrode. In this state, the first end portion **118** of the plunger abuts the paper barrier **128** as pressurized air flows from the suction electrode through the portion **80** of the pneumatic valve **32** and into the valve body portion **84** as indicated by arrow **130** lifting the first end portion **118** of the plunger **116** away from a shoulder of the valve body portion **84** (i.e., actuating the pneumatic valve **32**). The pressurized air then flows through the valve body portion **84** past the second end portion **120** of the plunger **116** into the venturi path **34** and to the pump (and control board having the pressure sensor) as indicated by arrow **132**.

(41) FIG. **14** depicts the pneumatic valve **32** in the presence of a vacuum (e.g., triggered by pressing the suction electrode). In the presence of the vacuum or suction pressure from the pump, the first end portion **118** of the plunger **116** abuts the shoulder **134** of the valve body portion **84** causing suction pressure on the suction electrode.

(42) FIG. **15** is a side view of the electrode distributor **12** coupled to the lead wires **38**. As depicted, the lead wires **38** of the suction electrodes are coupled to the bottom surface **106** of the housing **14** of the electrode distributor **12**. In particular, the p-connectors **108** are disposed in ports **136** in the

bottom surface **106**. The p-connectors **108** extend vertically in direction **64** from the bottom surface **106**. This minimizes or eliminates bending stresses in the lead wires **38**, thus, improving the life span of the lead wires **38**.

(43) In addition, the electrode distributor **12** includes a holder **138** for receiving and coupling an acquisition module (e.g., acquisition module **42** in FIG. **1**) to the electrode distributor **12**. The holder **138** is disposed on a top surface (e.g., the lid **100**) of the electrode distributor **112**.

(44) FIG. **16** is a perspective view of the ECG electrode application system **10**. As depicted, the acquisition module **42** is disposed within the holder **138** on top of the electrode distributor **12**. Also, depicted in FIG. **16**, an outer surface **140** of the regulator **26** includes a knob **142** (e.g., slider) that may be displaced between sides **29**, **31** to adjust the vacuum.

(45) The ECG electrode application system **10** includes a more compact configuration than other ECG electrode application systems with the acquisition module **42** disposed directly over the electrode distributor **12** and the regulator **26**. The electrode distributor **12** coupled to the regulator **12** includes a first dimension **144** (e.g., length), a second dimension **146** (e.g., width), and a third dimension **148** (e.g., height). Besides being more compact, the ECG electrode application system **10** is easily cleaned.

(46) Technical effects of the disclosed aspects include providing an ECG electrode application system including an electrode distributor coupled to suction electrodes. The electrode distributor includes pneumatic valves integrated within a manifold disposed within the electrode distributor. Adjacent pneumatic valves may be arranged in both a first direction and a second direction (which are orthogonal with respect to each other) along a horizontal plane. Each pneumatic valve includes a venturi path integrated within the manifold. The venturi path determines the negative pressure (e.g., suction dome pressure) with which the suction electrode is adhered to the skin of a patient. An orifice is disposed with the common manifold chamber. The orifice determines the sensitivity (e.g., by controlling mass flow rate within the system) for pressing suction dome to trigger a pump (e.g., disposed within the electrode distributor) to cause a vacuum to create the negative pressure for adhering the suction electrodes to the skin of the patient. The utilization of the venturi paths enables a compact manifold. The combination of the venturi paths and the orifice improves triggering the use of a pump and provides sufficient vacuum transfer pressure to the suction electrodes.

(47) The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . . ” or “step for [perform]ing [a function] . . . ”, it is intended that such elements are to be interpreted under 35 U.S.C. 112 (f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112 (f).

(48) This written description uses examples to disclose the present subject matter, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the subject matter is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. An electrocardiogram (ECG) electrode application system, comprising: an electrode distributor, comprising: a housing; and a suction manifold assembly disposed within the housing, wherein the suction manifold assembly comprises a manifold, a plurality of pneumatic valves integrated within the manifold, and the manifold comprises a common manifold chamber fluidly coupled to each pneumatic valve of the plurality of pneumatic valves, and each pneumatic valve comprises a venturi passage, wherein each pneumatic valve comprises a valve body portion coupled to an inlet portion, and each pneumatic valve comprises an outlet portion and a narrow portion disposed between the valve body portion and the outlet portion, and wherein the narrow portion has a diameter that is less than a respective diameter of each of the inlet portion, the valve body portion, and the outlet portion; and a plurality of suction electrodes, each suction electrode of the plurality of suction electrodes having an electrode suction dome and coupled to the electrode distributor via a lead wire, and each lead wire is coupled to a respective pneumatic valve; wherein each electrode suction dome is configured, when pressed to cause a change in air pressure that results in a suction pressure being applied within a respective electrode suction dome, to adhere a corresponding suction electrode to a subject at a pressure determined by the venturi passage of a corresponding pneumatic valve.
2. The ECG electrode application system of claim 1, further comprising a pump disposed within the housing, wherein the pump is configured to generate a vacuum to create the suction pressure when the respective electrode suction dome is pressed.
3. The ECG electrode application system of claim 1, further comprising an orifice disposed within the common manifold chamber, wherein the orifice is configured to set a level of change in the air pressure to trigger the application of the suction pressure applied within the respective electrode suction dome.
4. The ECG electrode application system of claim 3, further comprising a regulator configured to regulate a vacuum utilized to create the suction pressure, wherein the regulator is configured to couple to the housing, and the orifice is coupled to the regulator.
5. The ECG electrode application system of claim 1, wherein the housing comprises a first surface and a second surface disposed opposite the first surface, wherein the first surface is configured to receive an acquisition module for acquiring ECG signals from the plurality of suction electrodes and the second surface is configured to receive the respective lead wires of the plurality of suction electrodes.
6. The ECG electrode application system of claim 5, wherein the first surface is a top surface and the second surface is a bottom surface.
7. The ECG electrode application system of claim 1, wherein pneumatic valves of the plurality of pneumatic valves are disposed adjacent to each other along a horizontal plane in both a first direction and a second direction, the first direction being orthogonal to the second direction.
8. The ECG electrode application system of claim 7, wherein the plurality of pneumatic valves comprises both a first set of pneumatic valves disposed along a first horizontal plane in both the first direction and the second direction and a second set of pneumatic valves disposed along a second horizontal plane in both the first direction and the second direction, the first horizontal plane being parallel with the second horizontal plane.
9. The ECG electrode application system of claim 8, wherein a number of pneumatic valves in the first set of pneumatic valves is different from a number of pneumatic valves in the second set of pneumatic valves.
10. The ECG electrode application system of claim 9, wherein the first set of pneumatic valves comprises 6 pneumatic valves and the second set of pneumatic valves comprises 8 pneumatic valves.
11. The ECG electrode application system of claim 8, wherein the first set of pneumatic valves is oriented in an opposite direction from the second set of pneumatic valves.

12. The ECG electrode application system of claim 8, wherein the first set of pneumatic valves is disposed above the second set of pneumatic valves in a vertical direction that is orthogonal to the first and second horizontal planes.
13. The ECG electrode application system of claim 1, wherein each venturi passage is integral to the manifold.
14. The ECG electrode application system of claim 13, wherein the manifold is additively manufactured.
15. A suction manifold assembly for an electrocardiogram (ECG) electrode application system, comprising: a manifold; a plurality of pneumatic valves integrated within the manifold; a common manifold chamber fluidly coupled to each pneumatic valve of the plurality of pneumatic valves, and wherein each pneumatic valve is configured to be coupled via a lead wire to a respective suction electrode; wherein the suction manifold assembly is configured to be disposed within a housing of an electrode distributor of the ECG application system, and wherein pneumatic valves of the plurality of pneumatic valves are disposed adjacent to each other along a horizontal plane in both a first direction and a second direction, the first direction being orthogonal to the second direction; and wherein the plurality of pneumatic valves comprises both a first set of pneumatic valves disposed along a first horizontal plane in both the first direction and the second direction and a second set of pneumatic valves disposed along a second horizontal plane in both the first direction and the second direction, the first horizontal plane being parallel with the second horizontal plane, and wherein the first set of pneumatic valves is disposed in an opposite orientation from the second set of pneumatic valves.
16. The ECG electrode application system of claim 15, wherein a number of pneumatic valves in the first set of pneumatic valves is different from a number of pneumatic valves in the second set of pneumatic valves.
17. The ECG electrode application system of claim 15, wherein the first set of pneumatic valves is disposed above the second set of pneumatic valves in a vertical direction that is orthogonal to the first and second horizontal planes.
18. An electrocardiogram (ECG) electrode application system, comprising: an electrode distributor, comprising: a housing; and a suction manifold assembly disposed within the housing, wherein the suction manifold assembly comprises a manifold, a plurality of pneumatic valves integrated within the manifold, and the manifold comprises a common manifold chamber fluidly coupled to each pneumatic valve of the plurality of pneumatic valves, and each pneumatic valve comprises a venturi passage, wherein each pneumatic valve comprises a valve body portion coupled to an inlet portion, and each pneumatic valve comprises an outlet portion and a narrow portion disposed between the valve body portion and the outlet portion, and wherein the narrow portion has a diameter that is less than a respective diameter of each of the inlet portion, the valve body portion, and the outlet portion; an orifice disposed within the common manifold chamber; and a plurality of suction electrodes, each suction electrode of the plurality of suction electrodes having an electrode suction dome and coupled to the electrode distributor via a lead wire, and each lead wire is coupled to a respective pneumatic valve; wherein each electrode suction dome is configured, when pressed to cause a change in air pressure that results in a suction pressure being applied within a respective electrode suction dome, to adhere a corresponding suction electrode to a subject at a pressure determined by the venturi passage of a corresponding pneumatic valve, and the orifice is configured to set a level of change in the air pressure to trigger the application of the suction pressure applied within a respective electrode suction dome.